DEEP ANALYSIS ON DRUG OVERDOSE TRENDS IN THE US

By Jeremy Wu

Introduction

As a Seattle resident, witnessing the drug crisis firsthand has fueled my personal interest in understanding its patterns and origins. Coming from Asia, where public drug use is far less visible, this issue stood out immediately to me upon arriving. This analysis examines demographic risk factors — sex, race, and age — to identify which groups face the greatest risk of drug abuse and fatal overdose, with the hope of contributing meaningful insights toward addressing this public health crisis for a positive change!

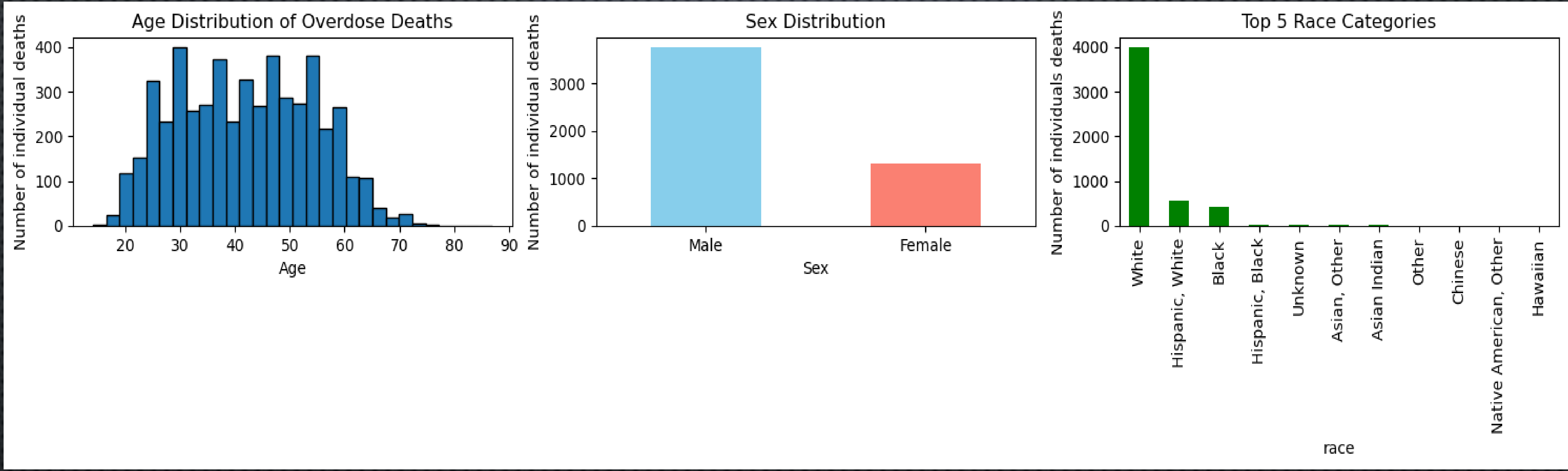
Goal: This project investigates a public dataset that links demographic, temporal and geographic patterns in Connecticut to drug overdose deaths (2012-2017). Using an appropriate predictive model, I will try to predict fentanyl involvement based on certain features/characteristics.

Data Source

- This data came from Kaggle
- The dataset has 42 columns, 5105 rows Total
- Key Variables (columns)
  - Date – date when the individual passed
  - Age
  - Sex
  - Race
  - Residence County
  - Death County
  - COD – Cause of death (which drug type, or multiple drugs)
  - Heroin – 1 represents present in the body during test, 0 represents not detected in the body when tested after death
  - Cocaine – 0 & 1
  - Fentanyl – 0 & 1

Distribution of Age, Sex, and Race:

The datasets distribution is depicted below in terms of age, sex, and race. This provides a grouping perspective of individuals that overdosed in Connecticut and the patterns/likelihood of individuals overdosing on drugs based on these 3 categories.



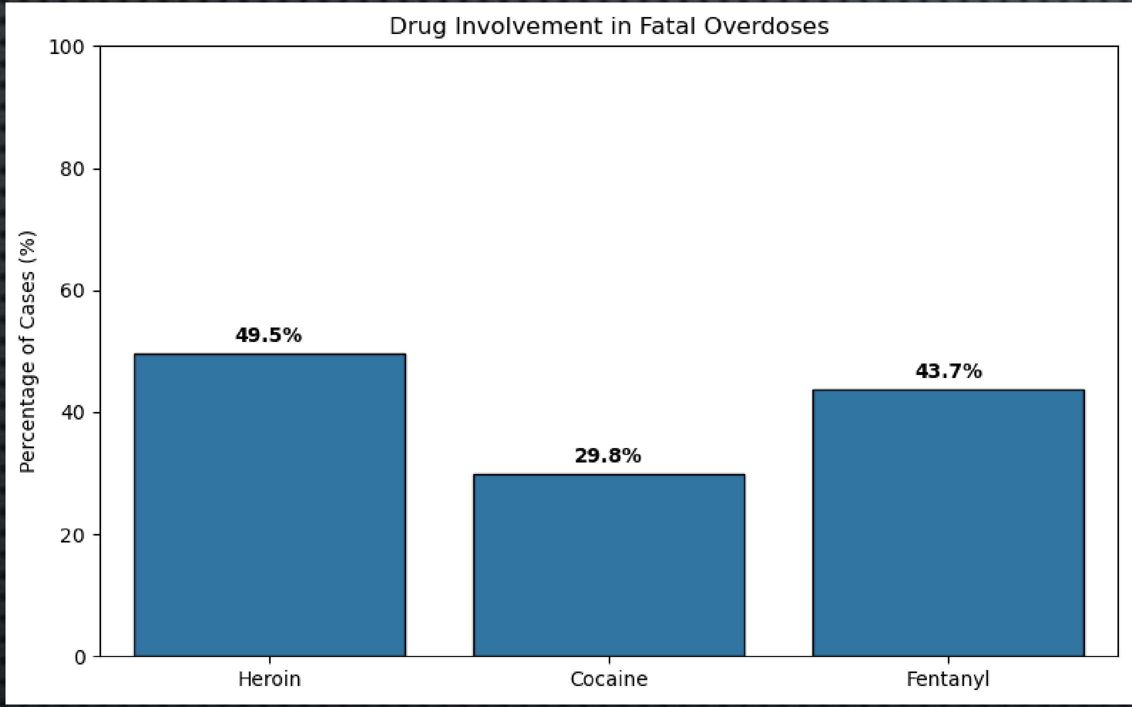
The population in Connecticut between 2012-2017 is approximately 3,590,000. Out of this:

1. 71-73% were White, 10% were black, 11-14% Hispanic(Latin America, with some overlaps in other races), and 4-5% were Asian
2. Males and Females were equally split around 49% Male and 51% Female
3. Connecticut had a median age of 41 years old, with majority individuals between 30-64 years old

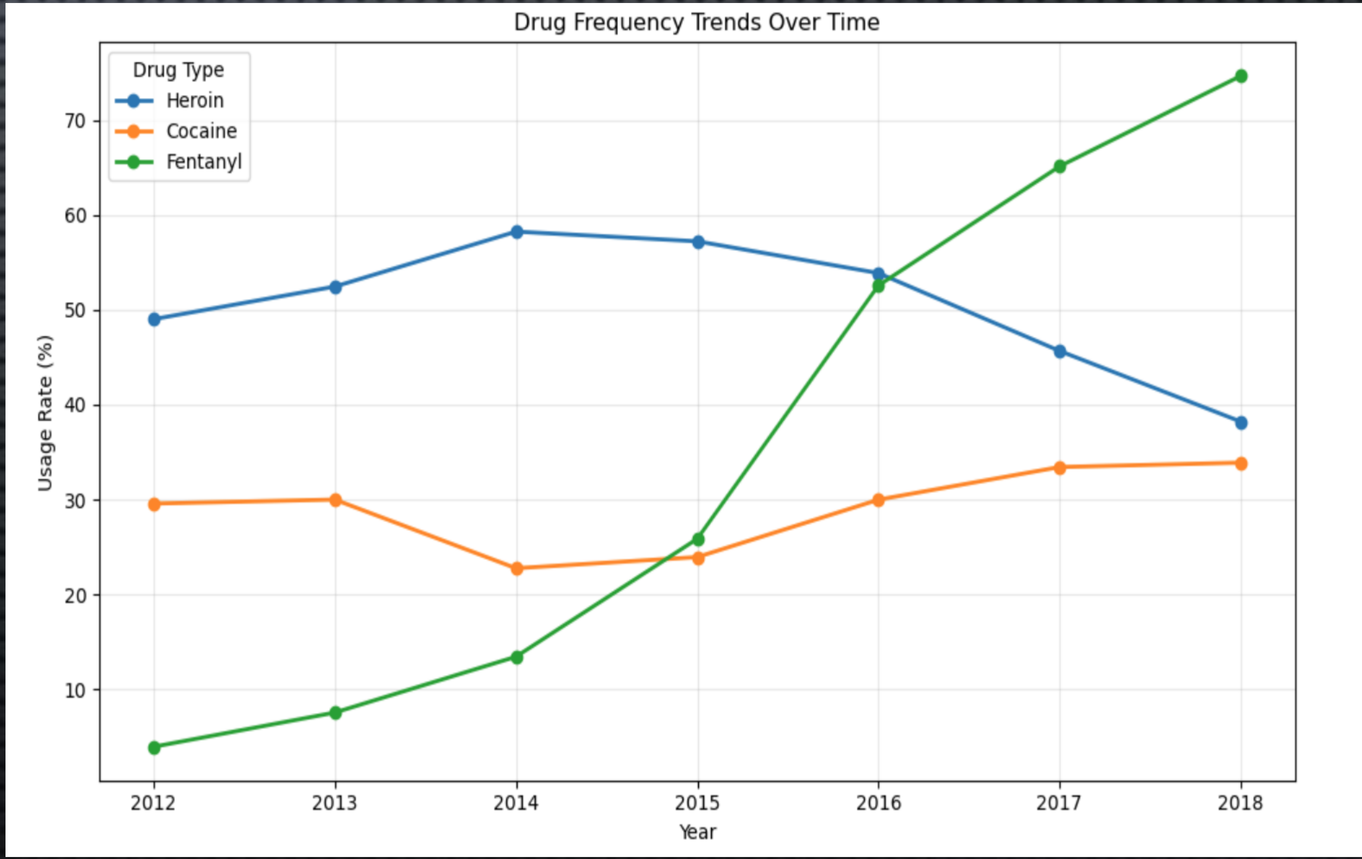
SPECIFIC DRUG TYPE THAT CAUSED OVERDOSE AND ITS USAGE TREND OVER TIME

- Heroin & Fentanyl is detected significantly higher for an individual who overdosed compared to Cocaine

Cocaine is more expensive and more difficult accessing, whilst fentanyl is so overly produced and easily accessible, we see the results of accessibility and cost of drug here.



- The drug that has the largest impact on US individuals: Fentanyl
  - sharp increase in drug frequency since 2012, with the largest increase in 2015 (steepest slope), and it is not slowing down anytime soon
  - In 2018, Fentanyl reached an all-time high of 74% likelihood of detection in an individual who passed from drug overdose. That means almost 3/4 drug overdose individuals had fentanyl detected in their body.
- Heroin and Cocaine both are steadier in terms of its frequency from 2012-2018, suggesting similar number of passings over the past 6 years from these 2 drugs.



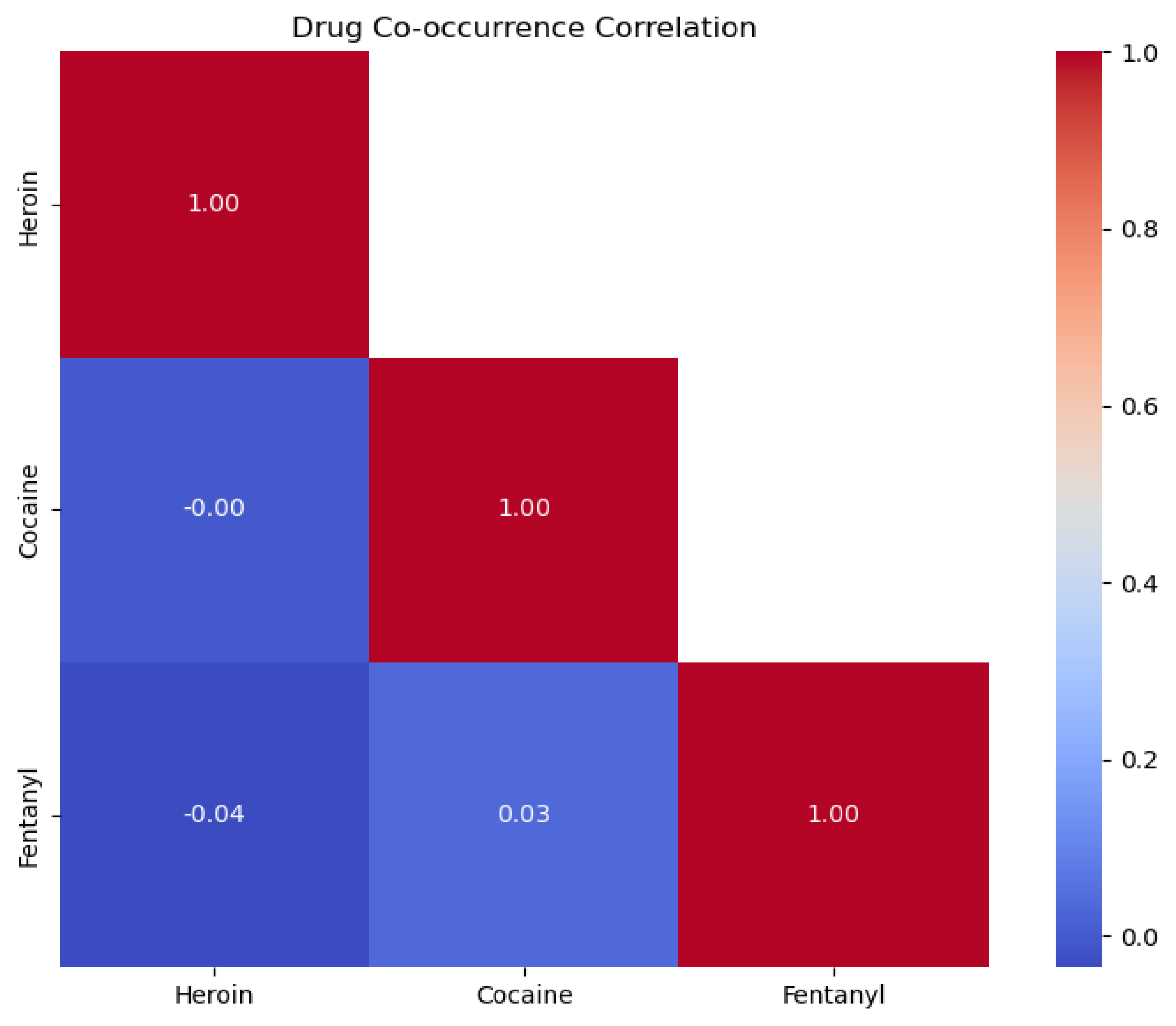
SPECIFIC DRUG TYPE AND IT'S CORRELATION WITH EACH OTHER

What are the chances of finding another drug present when [Heroin/Cocaine/Fentanyl] is detected?

There is no correlation between heroin and cocaine, however, there is a correlation with both drugs when Fentanyl is detected (cocaine and heroin), suggesting Fentanyl users are more prone to using other drugs at the same time, while individuals who use cocaine are less likely to mix drugs at the same time.

This poses the question:

1. Why do individuals who do Fentanyl mix it with other drugs?
2. Would eliminating Fentanyl also eliminate other drug uses?



COUNTY & OVERDOSE LEVELS

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=== Drug Involvement in Top 5 Counties

HARTFORD (n=1233):
  Heroin: 44.0%
  Cocaine: 31.8%
  Fentanyl: 48.2%

NEW HAVEN (n=1107):
  Heroin: 51.6%
  Cocaine: 32.1%
  Fentanyl: 35.9%

FAIRFIELD (n=623):
  Heroin: 48.8%
  Cocaine: 30.7%
  Fentanyl: 43.8%

NEW LONDON (n=368):
  Heroin: 50.0%
  Cocaine: 24.5%
  Fentanyl: 39.7%

LITCHFIELD (n=237):
  Heroin: 58.2%
  Cocaine: 22.4%
  Fentanyl: 40.9%

MIDDLESEX (n=181):
  Heroin: 42.5%
  Cocaine: 28.7%
  Fentanyl: 43.6%

WINDHAM (n=142):
  Heroin: 52.1%
  Cocaine: 23.9%
  Fentanyl: 46.5%

TOLLAND (n=113):
  Heroin: 38.9%
  Cocaine: 25.7%
  Fentanyl: 51.3%
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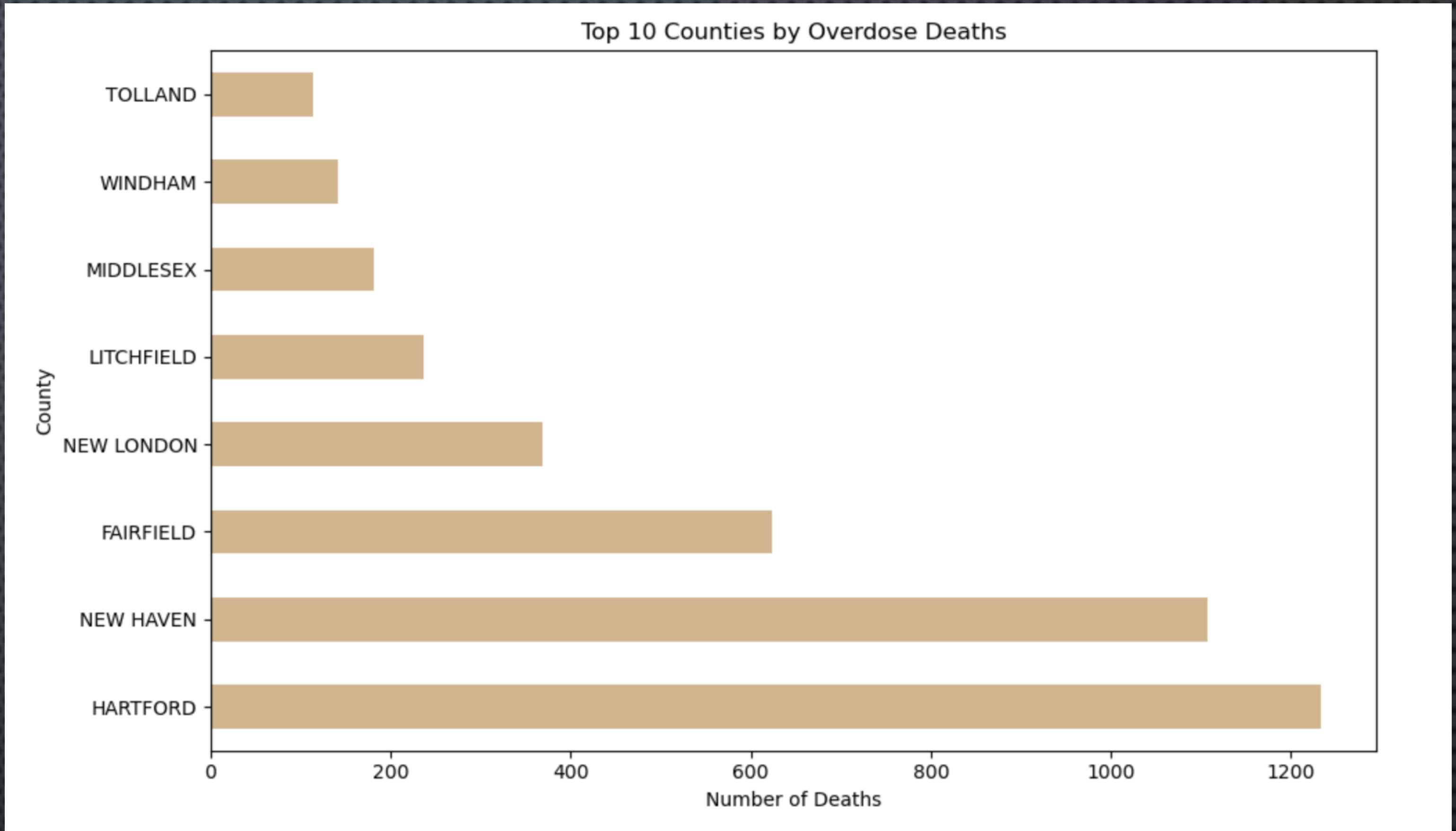

Each county is ordered with "total number of overdose deaths":

We found Hartford as the county with the most overdose deaths between 2012-2018, and Tolland county with the least number of deaths. In Hartford, fentanyl contributed to 48% of the deaths, whilst heroin contributed to 44% of deaths. In Tolland, 51% of deaths had fentanyl detected present, again suggesting its wide access and impact. The percentages do not add up to 100 because multiple drugs can be detected in an individual!

PREDICTIVE MODEL: RANDOM FOREST DEEP DIVE INTO FENTANYL

Target Variable: Fentanyl Present

- 1 = Fentanyl Present
- 0 = Fentanyl not present

Features used to predict presence of Fentanyl: Year, Age, Race, Sex, County, Cocaine (present or not), Heroin (present or not)

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=== Train/Test Split ===

Training set: 4084 samples (80.0%)
Test set: 1021 samples (20.0%)
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My model is trained with 80% of my dataset and 20% is tested for my model

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One-Hot Encoding complete:

Sex encoded into 2 binary columns:

   Columns: ['Sex_Male', 'Sex_Unknown']

Race encoded into 10 binary columns:

   Columns: ['Race_Asian, Other', 'Race_Black',
'Race_Chinese', 'Race_Hawaiian', 'Race_Hispanic, Black',
'Race_Hispanic, White', 'Race_Native American, Other',
'Race_Other', 'Race_Unknown', 'Race_White']

DeathCounty encoded into 8 binary columns:

   Columns: ['DeathCounty_HARTFORD',
'DeathCounty_LITCHFIELD', 'DeathCounty_MIDDLESEX',
'DeathCounty_NEW HAVEN', 'DeathCounty_NEW LONDON',
'DeathCounty_TOLLAND', 'DeathCounty_USA',
'DeathCounty_WINDHAM']
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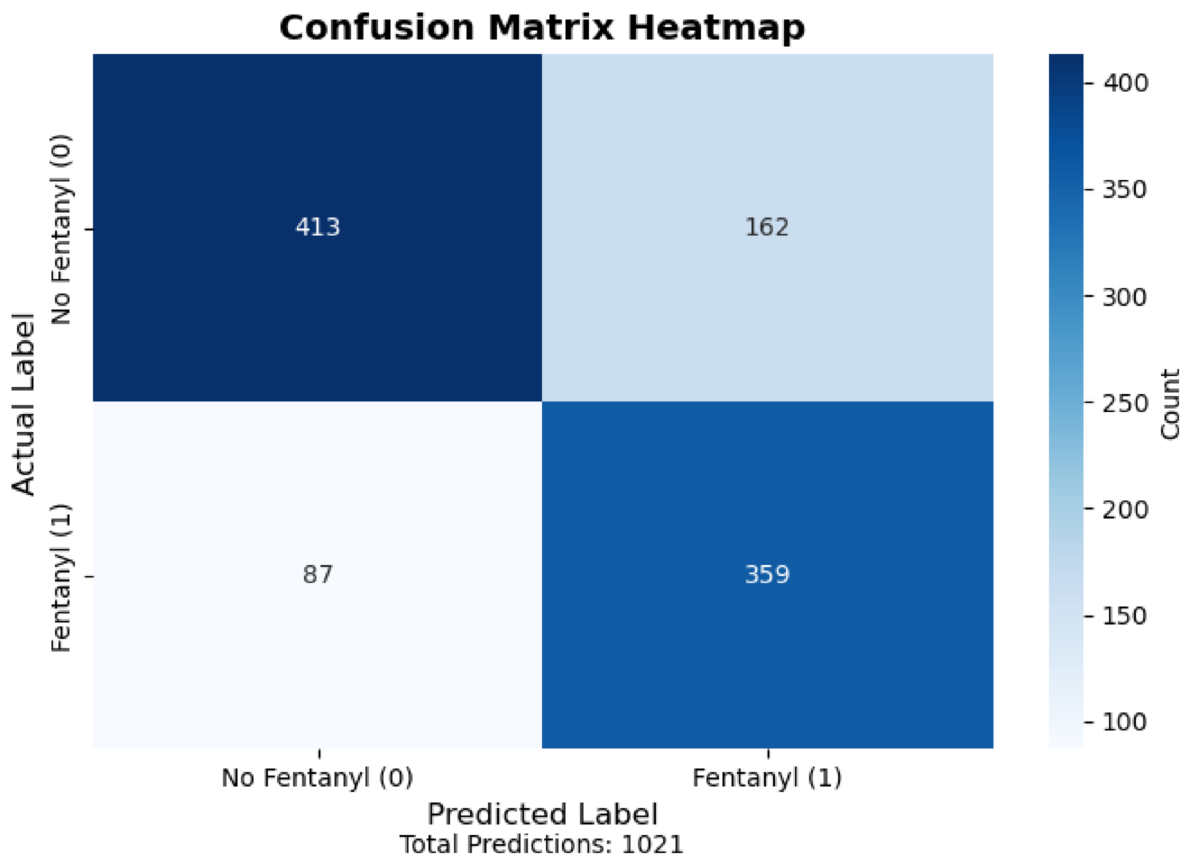
The values before I encoded the features: sex, race and death county were text strings. I used one-hot encoding to create a binary column for each unique value per feature (sex, race, death county) with a Boolean output (TRUE or FALSE)

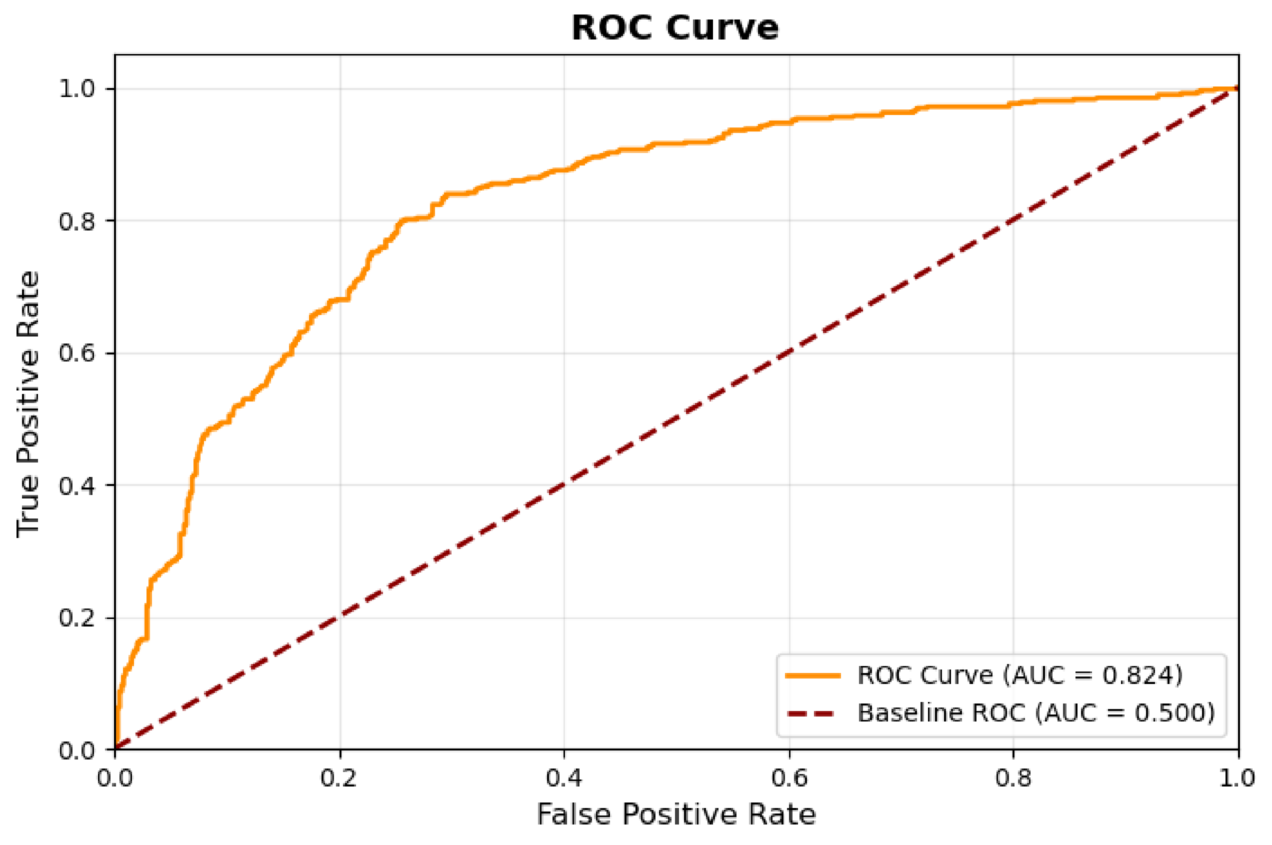
First 5 rows of new Death County Binary Columns:

DeathCounty_NEW HAVEN	DeathCounty_NEW LONDON	DeathCounty_TOLLAND
False	False	False
False	False	False
False	False	False
False	False	False
False	False	False

Race_Black	Race_Chinese	Race_Hawaiian	...	Race_Unknown	Race_White
False	False	False	...	False	False
True	False	False	...	False	False
False	False	False	...	False	True
False	False	False	...	False	True
False	False	False	...	False	False

First 5 rows of new Race Binary Columns:





My random forest model using the given 7 features to predict fentanyl activity resulted in an accuracy score of 75.6%. In addition, the F1 score 74.25% shows the balance between number of true positives compared to how many fentanyl cases we accurately predicted.

The Confusion Matrix Heatmap tells us we found 359 individuals accurately with fentanyl detected. We also accurately predicted 413 individuals with no fentanyl in their system out of 1021 total predictions, where the rest is false positives or false negatives.

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Performance & Accuracy Metrics:
  Accuracy:  0.7561 (75.61%)
  Precision: 0.6891 (68.91%)
  Recall:    0.8049 (80.49%)
  F1-Score:  0.7425 (74.25%)
  ROC-AUC:   0.8236 (82.36%)
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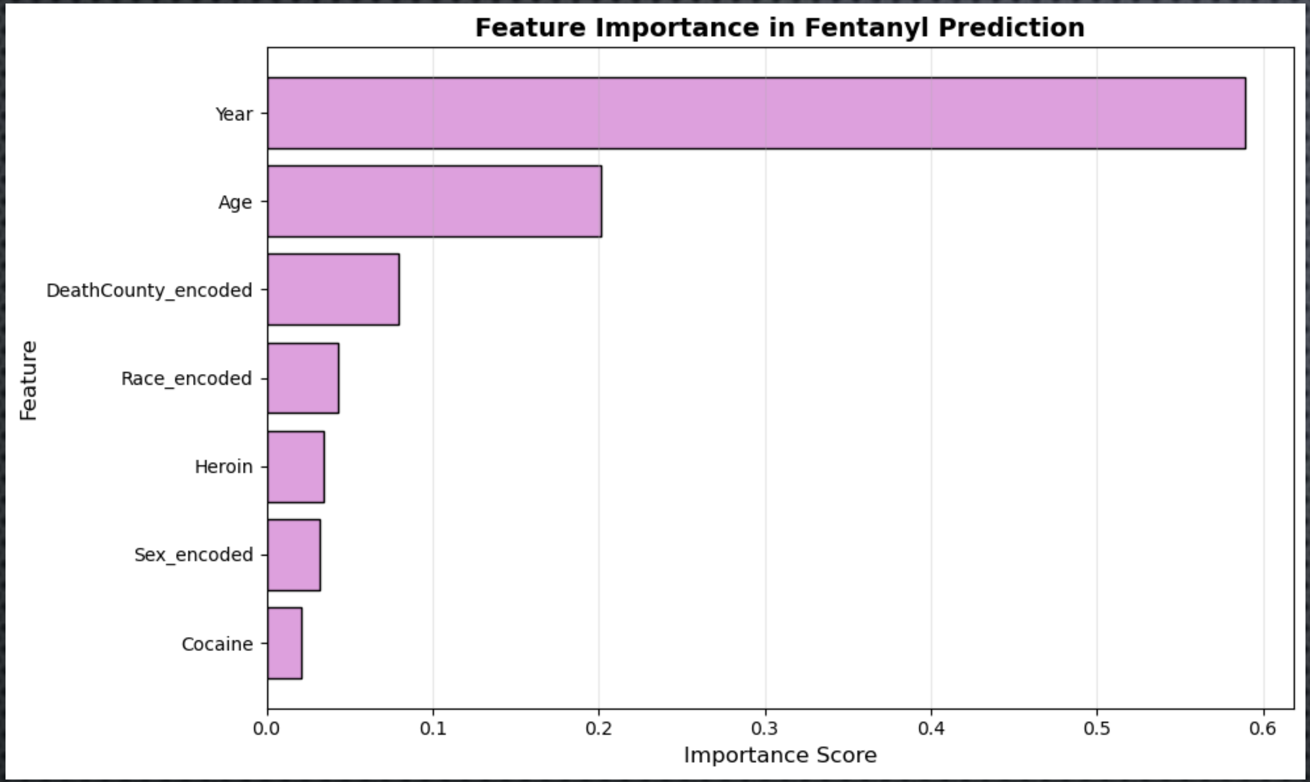

In my previous slide, The ROC curve for my random forest predictive model had an AUC (area under curve) score of 82.4%, suggesting that my model is accurate at predicting whether an individual will have fentanyl present in his/her body given all features. The baseline 45-degree diagonal line is a representation of an inaccurate prediction model

Feature Importance Scores:

Feature	Importance
Year	0.654487
Age	0.158643
Heroin	0.034999
Sex_Male	0.032825
Cocaine	0.018781
DeathCounty_HARTFORD	0.015560
DeathCounty_NEW HAVEN	0.014611
Race_White	0.011023
Race_Hispanic, White	0.009883
Race_Black	0.008427
DeathCounty_NEW LONDON	0.007227
DeathCounty_TOLLAND	0.006017
DeathCounty_LITCHFIELD	0.005677
DeathCounty_WINDHAM	0.005244
DeathCounty_MIDDLESEX	0.004504
Race_Hispanic, Black	0.003311
Race_Unknown	0.002724
Race_Asian, Other	0.002640
Race_Other	0.001966
Race_Chinese	0.000652
Race_Hawaiian	0.000442

Lastly, I calculated the importance of each individual feature in detecting fentanyl presence in overdosed individuals. Year is the most important feature on predicting fentanyl presence, further proving our point previously on the time trend line graph. In most recent years, fentanyl is more popular than back in 2012. After that, age is the second most important feature used to accurately predict fentanyl presence. Race has the lowest importance in our predictive model, perhaps due to the lack of race diversity in the population of Connecticut.

Conclusion & Interpretations

In conclusion, my analysis successfully demonstrated that fentanyl involvement in Connecticut overdose deaths can be predicted with high accuracy using a Random Forest classification model given the demographic, temporal, and drug interaction features. My model achieved high accuracy in identifying high-risk cases. Feature importance revealed that age and year were the dominant predictors, confirming the dramatic emergence of fentanyl over the past couple of years. My findings suggest that intervention strategies should prioritize Fentanyl & Heroin users and target specific age groups and geographic hotspots to combat this crisis. The presence of false negatives and false positives highlights the need for continued model refinement and other harm reduction approaches/features, including widespread fentanyl testing strip distribution particularly in high-risk counties.

Reference:

Drug overdose deaths. (n.d.). www.kaggle.com. https://www.kaggle.com/datasets/ruchi798/drug-overdose-deaths